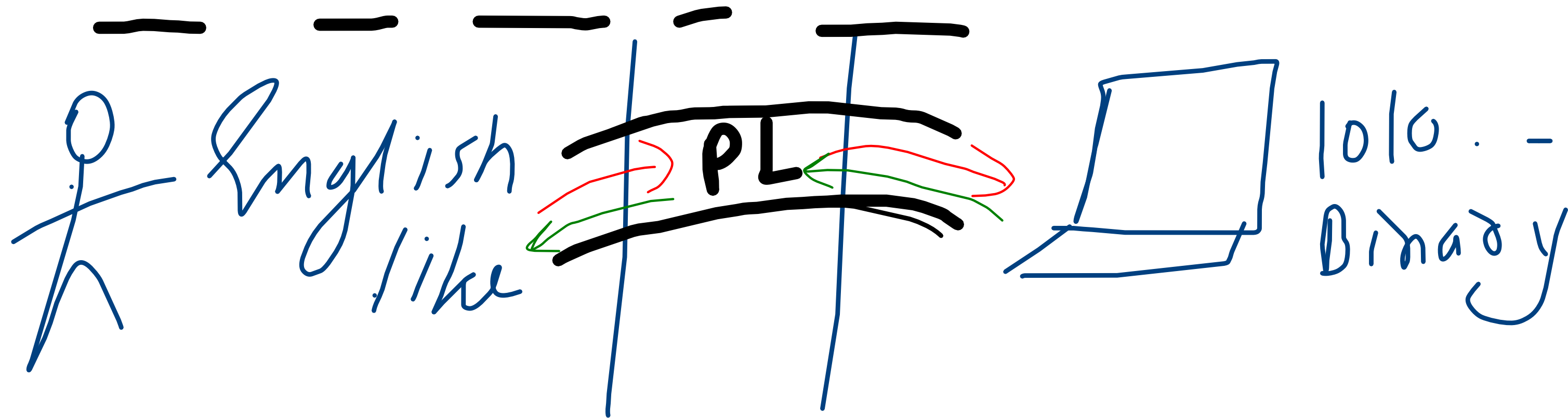




We have five generations of programming.

1 Machine language 101010101

2 Assembly language 1F,10

3 High-level language

**Maching independent + Supporting WORA architecture
gave being architecture neutral .**

**Why ?--->Compiler/Interpreter software:That converts high-level
language to machine-level language and converts the response of machine-
level language to high-level language (English).**

c/c++/java/python

4 Structured query language

SQL Language for CURD operation with reference to database.

**5 Natural language processing and AI
prompts**

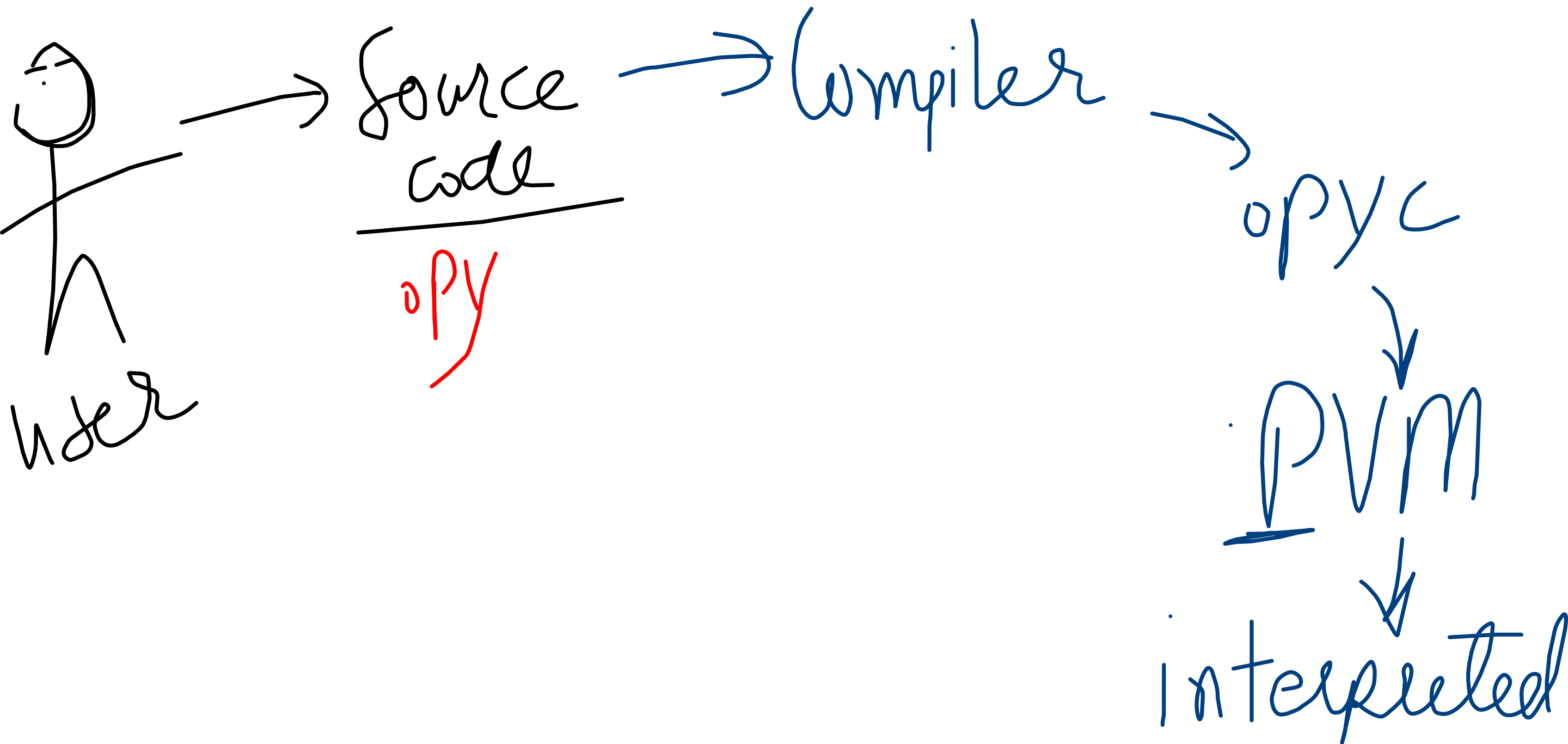
Dynamic casting ability to decide cast on basis of data.

a=10 (int)

a=a/3--->(float)

int,float,bool,str(string),complex

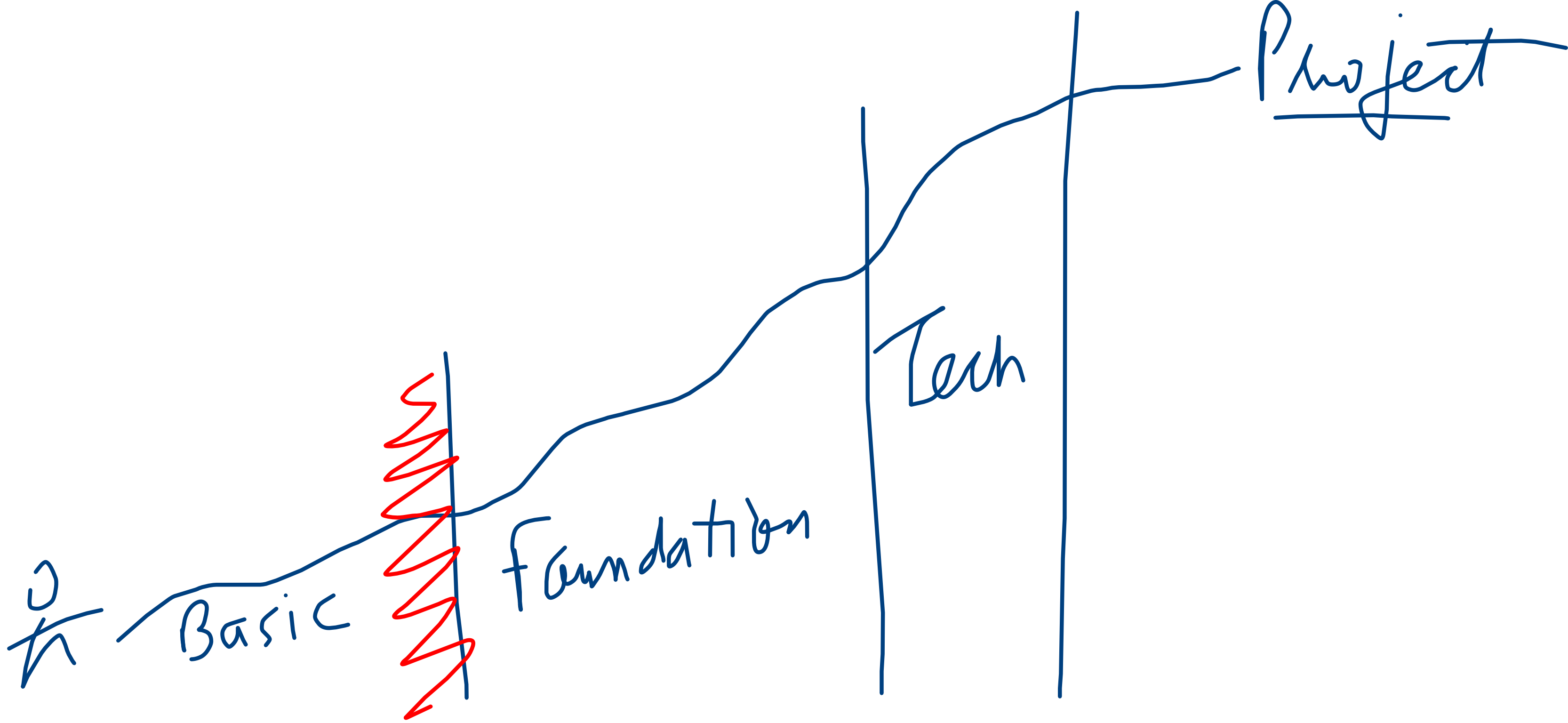
type()--->shows datatype casted to

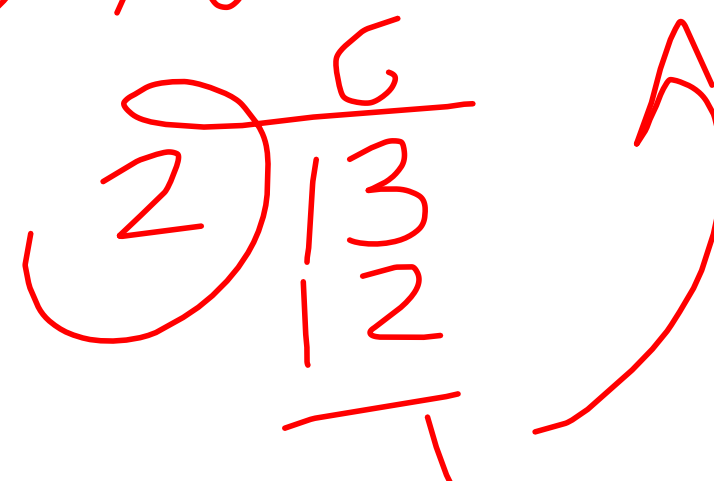


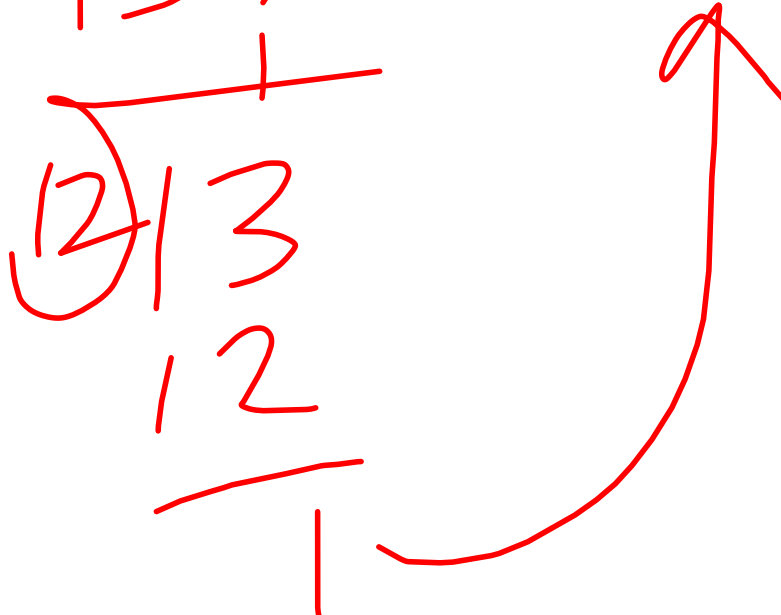
→ IDE → Python

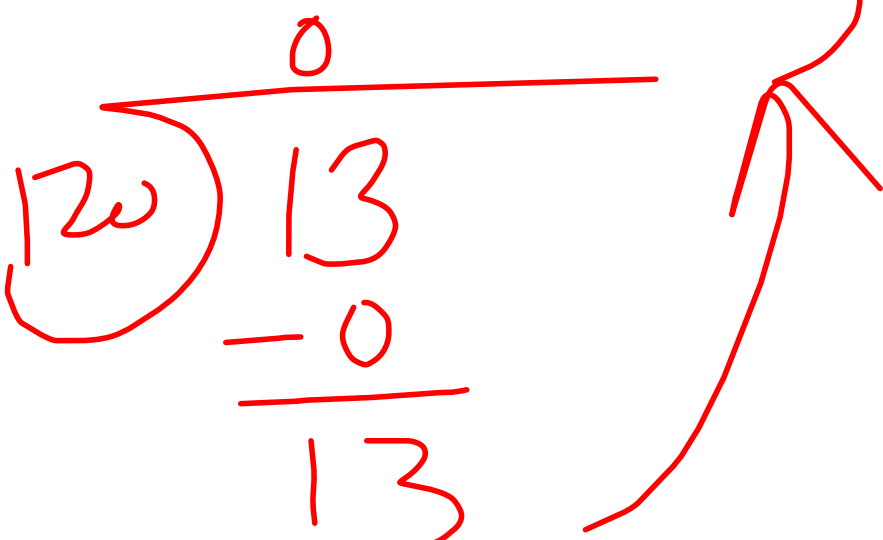
Python → PyCharm ✓
VS Code

→ Colab* Google



$$13 \% 2 = 1$$

$$\begin{array}{r} 6 \\ \hline 2 \overline{) 13} \\ \underline{12} \\ 1 \end{array}$$

$$13 \% 12 = 1$$

$$\begin{array}{r} \hline 12 \overline{) 13} \\ \underline{12} \\ 1 \end{array}$$

$$13 \% 20 = 13$$

$$\begin{array}{r} 0 \\ \hline 20 \overline{) 13} \\ \underline{0} \\ 13 \end{array}$$

$$2 * * 3 = 2^3 \Rightarrow 8$$

$$49 * * 0.5 = 7$$

1

$$\text{result} = 5 + 10 * 2 \rightarrow 5 + 20 \rightarrow 25$$

2

$$\text{result} = 50 \% 7 + 40 / 8 \rightarrow 1 + 5.0 = 6$$

3

$$\text{result} = (2 + 3) * (8 - 2) ** 2 \Rightarrow (5) * (6) ** 2$$
$$5 * 6 ** 2 = 5 * 36 = 180$$

4

$$\text{result} = -3 ** 2 + (4 - 6) * 2$$

5

$$\text{result} = (100 / 5) + 2 ** 3 * (7 \% 3) - 10$$

$$20 + 2 ** 3 * 1 - 10$$

$$20 + 8 * 1 - 10$$

$$20 + 8 - 10$$

$$28 - 10$$

$$18$$

$$- (3 ** 2) + (-2) * 2$$

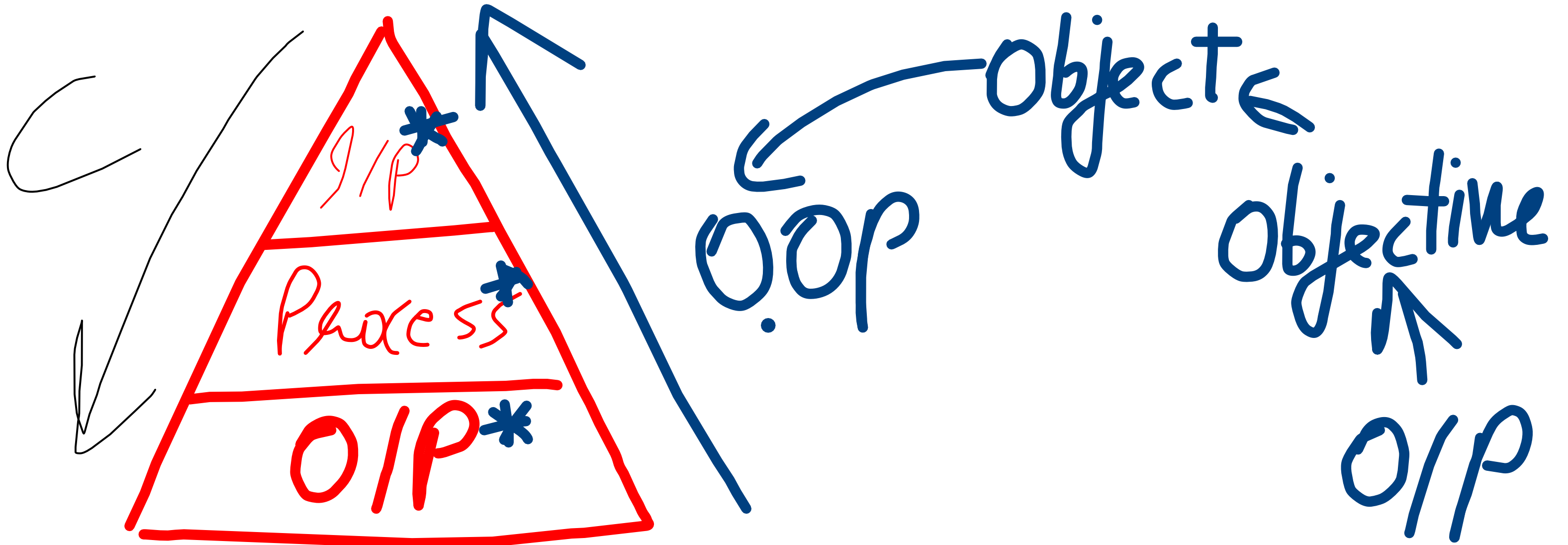
$$- 9 - 4$$

$$- 13 //$$

$$1428 \% 3000 = 1428$$



Write a program to convert Indian Rupees to US Dollars. 90inr:1usd



+ - * / // % **

=(assignment)

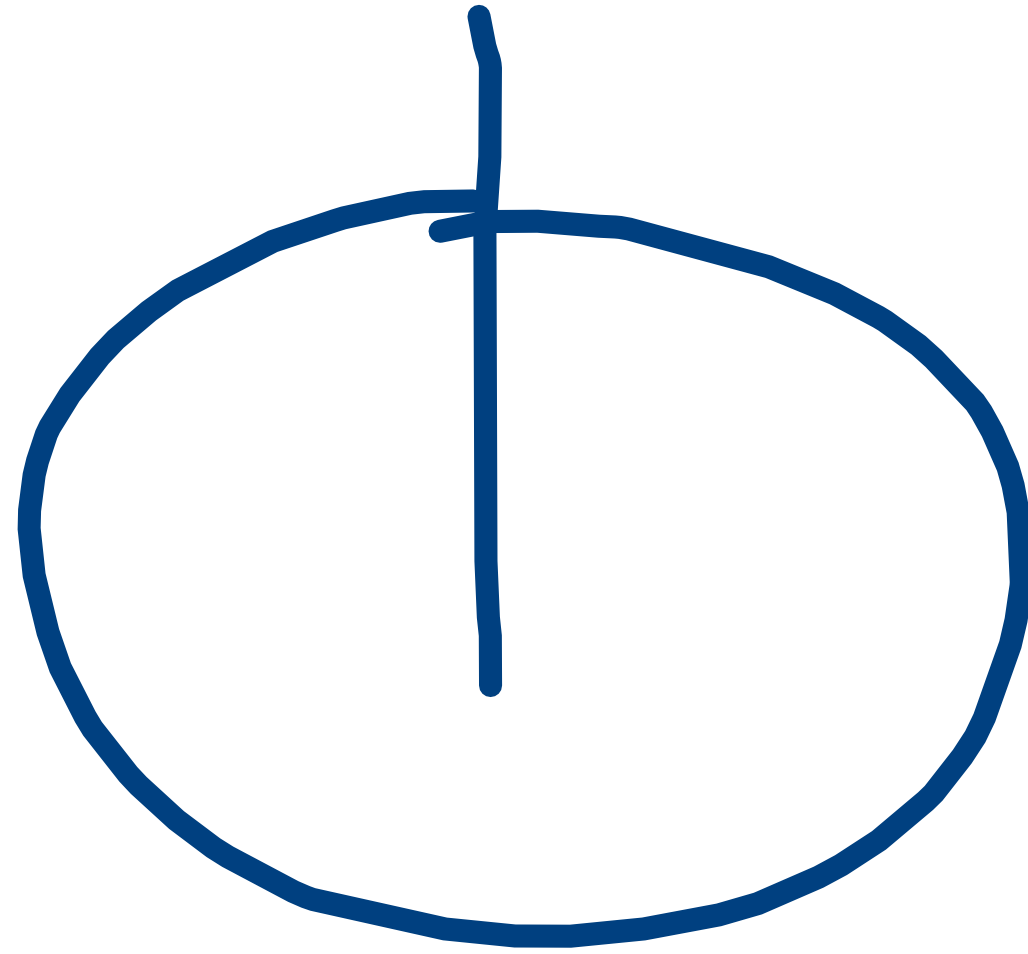
relational and logical

} True/False

65 53 1 42 3

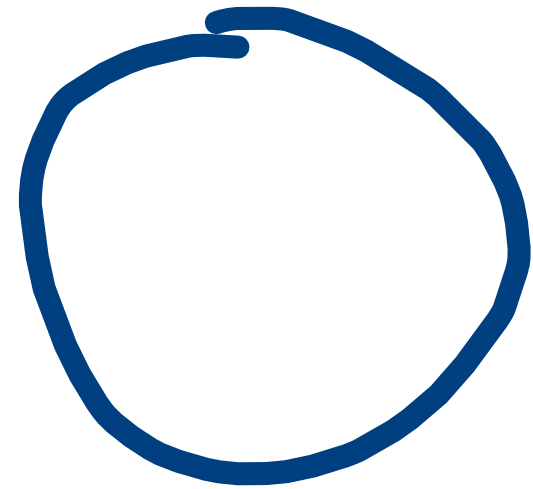
```
#Read passing percentage and print passing class,
#first class(>=60)
#second class(50<=p<60)
#third class(40<=p<50)
#fail(p<40)
percentage=float(input("Enter your percentage:"))
if percentage>=60:
    print(f"{percentage} is first class")
elif 50<=percentage<60:
    print(f"{percentage} is second class")
elif 40<=percentage<50:
    print(f"{percentage} is third class")
else:
    print(f"{percentage} is going for better luck next time")
```

exit



|

time
on



up


a,b,c=map(int,input("Enter 3 numbers:").split())

map
each
to
int

Read string

part
it
on
blank

'g'

dd,mm,yyyy = 6/10/2025

The problem definition for finding the roots of a quadratic equation is to **determine the values of the variable x that satisfy the equation $ax^2 + bx + c = 0$** , given the real coefficients a , b , and c , where $a \neq 0$.

The solution for the roots is given by the **quadratic formula**:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Conditions for Imaginary Roots

The nature of the roots (real or imaginary) is determined by the **discriminant** (Δ or D), which is the expression under the square root in the quadratic formula:

$$\Delta = b^2 - 4ac$$

< 0 then 'imag

Read a , b , c from the user using comma-separated values and produce root 1, root 2 if not imaginary.

Loop:
N number of instructions repeated in a fixed order for a finite time.
start: init value
step: +k(counter) -k(countdown)
 i<=n i>=0/1
end condition:where to stop

$n \rightarrow$ Product = 1 not 0
 $1 * 2 * 3 * 4$

for 1 to 4
1 to $n \rightarrow (1, n+1, 1)$

$n \rightarrow 1 * 2 * 3 * \dots$ factorial

for i in Range(1, $n+1$, 1)

Product $\times = i$

Product = Product $\times i$

$1 * 1$

$1 * 2$

$2 * 3$

$6 * 4 = 24$

~~1~~

Sum

Start

for i in range(1 7 + 1)

end

step

X
XX
XXX
XXXX
XXXXX
XXXXXX

X X X X X X
.

X X X X X

X X X

X X

X

for i in Range(6, 0, -1)

" " * (6-i), "x" * i

~~x~~~~x~~~~x~~~~x~~~~x~~~~x~~
~~x~~~~x~~~~x~~~~x~~~~x~~
~~x~~~~x~~~~x~~~~x~~~~x~~
~~x~~~~x~~~~x~~~~x~~
~~x~~~~x~~~~x~~

x	x	x	x	x	x
.	x	x	x	x	x
.	.	x	x	x	x
.	.	.	x	x	x
.	.	.	.	x	x
.	.	.	.	.	x

6
1.
2
3
4
5
5.
4
3
2
1

i=6

(6-i)

6

What program is made up of?

